Summary in Graph

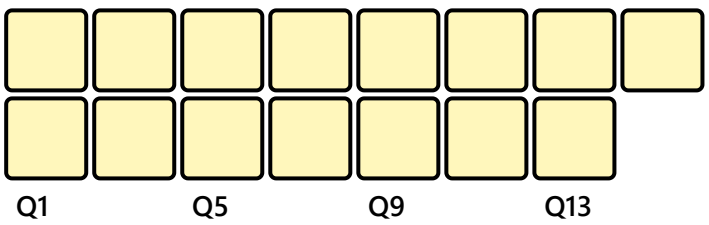
# Exam Summary (GO Classes GATE CS | Digital Logic | Number System | Weekly Quiz 3)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 15 (5 + 10) |
| Total Marks: | 25 (5 + 20) |
| Exam Duration: | 45 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (24) out of 235 users

EXAM RESPONSE EXAM STATS FEEDBACK

# Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Digital Logic

The Excess-$3$ code is also called

A. Cyclic Redundancy Code

B. Weighted Code

C. Self-Complementing Code

D. Algebraic Code

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following is the most efficient to perform arithmetic operations on the numbers?

A. Sign-magnitude

B. $1's$ complement

C. $2's$ complement

D. $9's$ complement

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

When two $n$-bit binary numbers are added the sum will contain at the most

A. $n$ bits

B. $n+2$ bits

C. $n+3$ bits

D. $n+1$ bits

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The Hexadecimal equivalent of $01111100110111100011$ is

A. CD73E

B. ABD3F

C. 7CDE3

D. FA4CD

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

If $12A7C_{16} = X_8$ then the value of $X$ is

A. $224174$

B. $425174$

C. $6173$

D. $225174$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Add the following numbers in $8$-bit $2's$ complement representation.
a) $10101101 + 110$
b) $01010111 + 011$
Which of the above operations result in an overflow ?

A. Only a
B. Only b
C. Both
D. None

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the answers below corresponds to the sum and overflow result when the following addition is performed using $8$-bit $2's$ complement: $10101101 + 110$?

A. $10101011$, with no overflow

B. $10101011$, with overflow

C. $10110011$, with no overflow

D. $10110011$, with overflow

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

In the decimal system with digits in the set $\{0, \ldots, 9\}$, a notation like $234$ stands for $2 \times 10^2 + 3 \times 10 + 4$; but in a different base $r$ with digits in the set $\{0, \ldots, r-1\}$, the notation $234$ will stand for the number $2 \times r^2 + 3 \times r + 4$. If the relation $430 + 240 = 1000$ holds when numbers are represented in a certain base $r$, then $r$ equals

A. $5$

B. $6$

C. $7$

D. $8$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Let $N = 2^{150}$. How many bits are required to write $N$ in binary (base 2)?

Your Answer: | Correct Answer: 151 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Let $N = 2^{150}$. How many digits are required to write $N$ in decimal (base 10)? You may assume that $\log_{10} 2 = 0.30103$.

Your Answer: | Correct Answer: 46 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

If $(12x)_3 = (123)_x$, then the value of $x$ is

A. $3$

B. $3$ or $4$

C. $2$

D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

If $73_x$ (in base $x$ number system) is equal to $54_y$ (in base $y$ number system), the possible values of $x$ and $y$ are

A. $8$ and $16$

B. $10$ and $12$

C. $9$ and $13$

D. $8$ and $11$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

One approach to handling fuzzy logic data might be to design a computer using ternary (base-$3$) logic so that data could be stored as “true,” “false,” and “unknown.” If each ternary logic element is called a flit, how many flits are required to represent at least $256$ different values?

A. $4$

B. $5$

C. $6$

D. $7$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

How many different $\mathrm{BCD}$ numbers can be stored in $12$ switches ? (Assume two position or on-off switches).

A. $2^{12}$

B. $2^{12} - 1$

C. $10^{12}$

D. $10^{3}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

If $N^2 = (7601)_8$ where $N$ is a positive integer, then the value of $N$ is

A. $(241)_5$

B. $(143)_6$

C. $(165)_7$

D. $(39)_{16}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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